Self-Assessment

AdventureWorks Power BI Trainee Project

Skill Ratings

Rated against the project rubric: Beginner / Developing / Proficient / Strong.

- Data modelling and relationships – Strong
- DAX measures and time intelligence – Strong
- Power Query and data transformation – Proficient
- Report design and UX – Proficient
- Row-level security – Developing
- Performance optimization – Proficient
- Documentation and communication – Strong

Row-level security is the one I'm rating Developing instead of Proficient. The static roles work fine, I tested them with View As and they filter correctly. But the dynamic RLS role uses USERPRINCIPALNAME(), and that function just doesn't return anything useful in Desktop - it only works once you've published to the Service and you're on Pro, which I'm not. So I wrote it, I'm fairly sure the logic is right, but I've never actually seen it work. Didn't feel right calling that Proficient when I can't prove it does what it's supposed to.

The Hardest Thing I Built

Probably the Power Query merge from Week 4. I merged Product and Budget to get a Budget Status column going, and I set it up with Budget as the base table and Product merged into it. Looked completely normal - the column showed up, numbers looked reasonable, moved on. Didn't think twice about it.

Except it had quietly broken my TREATAS measure from Week 2. Since a bunch of products share the same category, merging Product into Budget duplicated every Budget row - one copy per matching product instead of one row per category. That killed the row uniqueness on Budget[Category], which is the exact thing Budget Amount by Category needs to filter properly through TREATAS. And none of this threw an error. It just sat there being wrong.

I only caught it because I went back later to double check the TREATAS numbers against the raw Budget table, mostly out of paranoia, and they didn't add up. Took a while to actually trace it back to the merge direction, because that's not usually where I'd look first. Once I figured out what was going on the fix was easy - just flip it so Product is the base and Budget's amount gets pulled in as a lookup instead, which leaves Budget's rows alone.

The main thing I'm taking from it: which table you pick as the base query in a merge isn't just a formatting choice, it decides whose row count survives. If something downstream (like TREATAS) depends on one side staying unique, picking the wrong base breaks it silently instead of throwing an error you'd actually notice. I check for that now before merging anything.

What I Added After the First Submission

After the first version was done, I added a simulated Returns table and rebuilt the layout of every page. AdventureWorks has no returns data, so I made one in Power Query: it picks a random 8% of the sales lines and treats them as returned, with a random reason, a random return date and a return quantity. On top of that I built four measures (Total Returns, Return Rate %, Return Value Lost and Average Days to Return) and a Returns Analysis page.

Two things I learned from it. First, my random picker had an off-by-one mistake, so one of the five return reasons never showed up and the chart only had four bars. Fixing it just meant changing the range in the M code, but it's a good reminder to look at what a chart is actually showing instead of trusting that the code did what I meant. Second, random data changes every time it refreshes, so I can't quote exact return numbers, and the README says clearly that none of the returns figures are real findings.

For the layout, I set one set of rules for every page: page navigator on top, KPI cards in a row, charts underneath, equal sizes and gaps, and one colour theme. Dragging things around by eye kept leaving gaps and overlaps, so I ended up typing exact sizes and positions for each visual. Moving visuals also put my Show Bar and Show Line bookmarks out of sync on Sales Analysis, so I had to fix which chart was visible in each one and update both bookmarks. I also hid the early test pages, because they made the report look unfinished.

The last lesson came from rechecking my business insights summary against the report. Several of my findings turned out to be wrong because I had taken numbers at face value. My year-over-year measure compared part of 2020 with several earlier years combined, so it showed a 58% drop when sales were actually up 71% on the same dates a year earlier, and I rewrote it to compare like with like. A placeholder customer called “[Not Applicable]” makes up about 73% of sales, and I had been treating it as a Gold-tier customer. I had also named the wrong region as the weakest one. I rewrote the summary using only figures I could trace back to the report. From now on I check what a number actually contains before I build a finding on it.